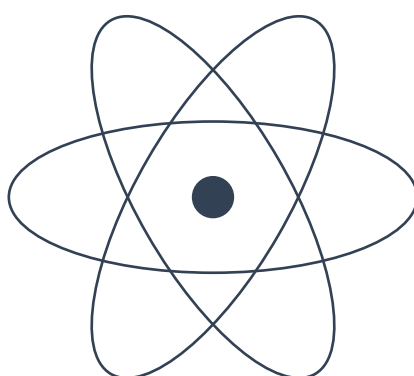


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Algebra I

Progressive Problem Set (40 Problems)

Olympiad Introductory Training • Math

SPhOH Team

FIRST EDITION • 2026

Contents

About This Problem Set	ii
1 Algebra I	1
1.1 Pure Algebraic Manipulations	1
1.2 Physical Algebra Applications	5
2 Hints	11

About This Problem Set

This problem set is part of the foundational training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It contains 40 structured problems arranged strictly by increasing difficulty, from single-step foundational algebra to challenging multi-step physics derivations.

Difficulty Levels

- **Level I** — Direct evaluations, single-step variable isolations, and standard formulas.
- **Level II** — Multi-step algebraic manipulations, systems of equations, and exponent rules.
- **Level III** — Coupled non-linear equations, approximations, and multi-variable physics derivations.

Find more at: <https://devfhd.github.io/SPhOH/>.

Algebra I

Topic Note. Mastering basic algebra is essential for physics. Problems are ordered sequentially from fundamental arithmetic operations to high-level system derivations and physical approximations.

1.1 Pure Algebraic Manipulations

PROBLEM 1

LEVEL I

Simple Linear Equation

Solve for x :

$$3x - 7 = 11$$

PROBLEM 2

LEVEL I

Pure Quadratic Equation

Solve for x :

$$2x^2 - 18 = 0$$

PROBLEM 3

LEVEL I

Simple Radical Equation

Solve for x :

$$\sqrt{x + 4} = 3$$

PROBLEM 4**LEVEL I****Simplifying Algebraic Fractions**

Simplify the fraction completely, assuming $x \neq 2$:

$$\frac{x^2 - 4}{x - 2}$$

PROBLEM 5**LEVEL I****Factoring Common Terms**

Factor completely:

$$3x^3 - 12x$$

PROBLEM 6**LEVEL I****Difference of Squares Mental Trick**

Calculate the value of $105^2 - 95^2$ without using a calculator.

PROBLEM 7**LEVEL I****Linear Inequality**

Solve for x and state the solution set:

$$-2x + 5 > 1$$

PROBLEM 8**LEVEL II****Factoring Quadratic Trinomial**

Solve for all real values of x :

$$x^2 - 5x + 6 = 0$$

PROBLEM 9**LEVEL II****System of Linear Equations**

Solve the system for x and y :

$$\begin{aligned}x + y &= 10 \\ 2x - y &= 5\end{aligned}$$

PROBLEM 10**LEVEL II****Variable Substitution**

Given $u = x - 1$, substitute u into the equation and solve for x :

$$(x - 1)^2 - 4 = 0$$

PROBLEM 11**LEVEL II****Exponents and Simplification**

Simplify the expression into the form 2^k :

$$\frac{2^5 \cdot 4^2}{8}$$

PROBLEM 12**LEVEL II****Proportionality Relations**

If y is directly proportional to x^2 and $y = 12$ when $x = 2$, find y when $x = 4$.

PROBLEM 13**LEVEL II****Clearing Fractional Denominators**

Solve for x :

$$\frac{1}{x} + \frac{1}{2x} = 1$$

PROBLEM 14**LEVEL II****Logarithmic Equation**

Solve for x :

$$\log_{10}(x) + \log_{10}(2) = 1$$

PROBLEM 15**LEVEL III****Completing the Square**

Rewrite $x^2 + 6x + 2 = 0$ in the form $(x + h)^2 = k$ and solve for x .

PROBLEM 16**LEVEL III****Exponential Quadratic Form**

Solve for x :

$$2^{2x} - 5 \cdot 2^x + 4 = 0$$

PROBLEM 17**LEVEL III****Nested Radical Equation**

Solve for all real x :

$$\sqrt{x+3} = x-3$$

PROBLEM 18**LEVEL III****Non-linear System of Equations**

Solve the following system for x and y assuming $x, y > 0$:

$$\begin{aligned}xy &= 12 \\ x^2 + y^2 &= 25\end{aligned}$$

PROBLEM 19**LEVEL III****Rational Expression Decomposition**

Express the fraction as a sum of partial fractions:

$$\frac{5x-1}{x^2-1}$$

PROBLEM 20**LEVEL III****Logarithmic System**

Solve for x and y :

$$\begin{aligned}\log_2(x) + \log_2(y) &= 3 \\ x - y &= 2\end{aligned}$$

1.2 Physical Algebra Applications

PROBLEM 21

LEVEL I

Constant Speed Relation

An object travels distance $d = vt$. If speed v is doubled while distance d stays constant, what happens to time t ?

PROBLEM 22

LEVEL I

Ideal Gas Temperature Isolation

The ideal gas law is $PV = nRT$. Isolate T in terms of P , V , n , and R .

PROBLEM 23

LEVEL I

Work Done by a Constant Force

Work is $W = Fd \cos(\theta)$. Find W if $F = 50 \text{ N}$, $d = 4 \text{ m}$, and $\theta = 60^\circ$ ($\cos(60^\circ) = 0.5$).

PROBLEM 24

LEVEL I

Thermal Heat Capacity

Heat energy is $Q = mc\Delta T$. Isolate ΔT in terms of Q , m , and c .

PROBLEM 25

LEVEL I

Hydrostatic Pressure Depth

Pressure at depth h is $P = P_0 + \rho gh$. Isolate depth h in terms of P , P_0 , ρ , and g .

PROBLEM 26

LEVEL I

Photon Energy Formula

Photon energy is $E = hf$. Substituting speed of light $c = f\lambda$, write E strictly in terms of h , c , and λ .

PROBLEM 27

LEVEL I

Rotational Energy Velocity

Rotational kinetic energy is $K_{\text{rot}} = \frac{1}{2}I\omega^2$. Isolate ω in terms of K_{rot} and I , assuming $I > 0$.

PROBLEM 28

LEVEL II

Kinematic Time Isolation

A car accelerates from rest ($v_0 = 0$) such that $d = \frac{1}{2}at^2$. Isolate time t in terms of d and a .

PROBLEM 29

LEVEL II

Ohm's Law Power Formula

Given $V = IR$ and $P = VI$, eliminate V to write power P in terms of I and R .

PROBLEM 30

LEVEL II

Resistors in Parallel

Two resistors R_1 and R_2 in parallel satisfy:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Solve explicitly for R_{eq} .

PROBLEM 31

LEVEL II

Gravitational Inverse-Square Law

Newton's law of gravitation is $F = G \frac{m_1 m_2}{r^2}$. If distance r is doubled, by what factor does force F change?

PROBLEM 32

LEVEL II

Conservation of Energy Speed

Equate potential energy and kinetic energy: $mgh = \frac{1}{2}mv^2$. Solve for v in terms of g and h .

PROBLEM 33

LEVEL II

Kinematic Distance Formula

Given $v^2 = v_0^2 + 2ad$, solve for displacement d in terms of v , v_0 , and a .

PROBLEM 34

LEVEL II

Spring Potential Energy Stretch

Spring energy is $U_s = \frac{1}{2}kx^2$. Isolate stretch x in terms of U_s and k , assuming $x > 0$.

PROBLEM 35

LEVEL III

Thin Lens Equation Isolation

The thin lens formula is:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Isolate image distance d_i in terms of f and d_o .

PROBLEM 36

LEVEL III

Centripetal Acceleration and Frequency

Centripetal acceleration is $a = \frac{v^2}{r}$ and speed is $v = 2\pi r f$. Substitute v into a to express acceleration in terms of π , r , and f .

PROBLEM 37

LEVEL III

Atwood Machine Acceleration

Solve the force system for acceleration a :

$$T - m_1 g = m_1 a$$

$$m_2 g - T = m_2 a$$

PROBLEM 38

LEVEL III

Pendulum Period Scaling

The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. If length L quadruples ($L' = 4L$), by what factor does T change?

PROBLEM 39

LEVEL III

Simple Binomial Approximation

For small $x \ll 1$, $(1 + x)^n \approx 1 + nx$. Use this to approximate $\frac{1}{1+x}$ when x is small.

PROBLEM 40

LEVEL III

Relativistic Kinetic Energy Limit

The relativistic energy equation is $E = \gamma mc^2$, where $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$. Express γ in terms of v and c , then use the binomial approximation $(1 - y)^{-1/2} \approx 1 + \frac{1}{2}y$ (for $y = v^2/c^2 \ll 1$) to show that kinetic energy $K = (E - mc^2)$ reduces to classical kinetic energy $\frac{1}{2}mv^2$.

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Simple Linear Equation

Add 7 to both sides, then divide by 3.

Hint 2 | Problem 2 – Pure Quadratic Equation

Add 18 to both sides, divide by 2 to get $x^2 = 9$, and take the square root. Include both positive and negative roots.

Hint 3 | Problem 3 – Simple Radical Equation

Square both sides of the equation to eliminate the square root.

Hint 4 | Problem 4 – Simplifying Algebraic Fractions

Factor the numerator using the difference of two squares: $x^2 - 4 = (x - 2)(x + 2)$.

Hint 5 | Problem 5 – Factoring Common Terms

First factor out $3x$, leaving $3x(x^2 - 4)$, then factor the difference of two squares.

Hint 6 | Problem 6 – Difference of Squares Mental Trick

Use the identity $a^2 - b^2 = (a - b)(a + b)$ with $a = 105$ and $b = 95$.

Hint 7 | Problem 7 – Linear Inequality

Subtract 5 from both sides, then divide by -2 . Remember to reverse the inequality sign when dividing by a negative number.

Hint 8 | Problem 8 – Factoring Quadratic Trinomial

Factor the quadratic into $(x - 2)(x - 3) = 0$, or use the quadratic formula.

Hint 9 | Problem 9 – System of Linear Equations

Add the two equations together to eliminate y directly and solve for x .

Hint 10 | Problem 10 – Variable Substitution

Rewrite as $u^2 - 4 = 0$, solve for $u = \pm 2$, then solve for $x = u + 1$.

Hint 11 | Problem 11 – Exponents and Simplification

Express all numbers with base 2: $4 = 2^2$ and $8 = 2^3$.

Hint 12 | Problem 12 – Proportionality Relations

Set up $y = kx^2$. Find $k = 12/2^2 = 3$, then evaluate $y = 3(4^2)$.

Hint 13 | Problem 13 – Clearing Fractional Denominators

Multiply the entire equation by $2x$ to clear all denominators.

Hint 14 | Problem 14 – Logarithmic Equation

Use the log product property $\log_{10}(2x) = 1$, then rewrite as $2x = 10^1$.

Hint 15 | Problem 15 – Completing the Square

Take half of the coefficient of x (3), square it (9), and add/subtract it to complete the square.

Hint 16 | Problem 16 – Exponential Quadratic Form

Let $u = 2^x$, turning the equation into $u^2 - 5u + 4 = 0$. Solve for u and then find x .

Hint 17 | Problem 17 – Nested Radical Equation

Square both sides to obtain $x + 3 = (x - 3)^2$. Solve the quadratic and check for extraneous roots.

Hint 18 | Problem 18 – Non-linear System of Equations

Use the identity $(x + y)^2 = x^2 + y^2 + 2xy$ to find $x + y$, then construct a quadratic equation.

Hint 19 | Problem 19 – Rational Expression Decomposition

Factor $x^2 - 1 = (x - 1)(x + 1)$ and set $\frac{A}{x-1} + \frac{B}{x+1} = \frac{5x-1}{x^2-1}$.

Hint 20 | Problem 20 – Logarithmic System

Convert the log equation to $xy = 2^3 = 8$, then substitute $x = y + 2$ into the product.

Hint 21 | Problem 21 – Constant Speed Relation

Substitute $v' = 2v$ into $d = v't'$ and solve for t' .

Hint 22 | Problem 22 – Ideal Gas Temperature Isolation

Divide both sides of the equation by nR .

Hint 23 | Problem 23 – Work Done by a Constant Force

Substitute $F = 50$, $d = 4$, and $\cos(60^\circ) = 0.5$ directly into the formula.

Hint 24 | Problem 24 – Thermal Heat Capacity

Divide both sides of the equation by mc .

Hint 25 | Problem 25 – Hydrostatic Pressure Depth

Subtract P_0 from both sides, then divide by ρg .

Hint 26 | Problem 26 – Photon Energy Formula

Substitute $f = \frac{c}{\lambda}$ into $E = hf$.

Hint 27 | Problem 27 – Rotational Energy Velocity

Multiply by 2, divide by I , and take the positive square root.

Hint 28 | Problem 28 – Kinematic Time Isolation

Multiply both sides by 2, divide by a , and take the positive square root.

Hint 29 | Problem 29 – Ohm's Law Power Formula

Substitute $V = IR$ directly into $P = VI$.

Hint 30 | Problem 30 – Resistors in Parallel

Find a common denominator R_1R_2 for the right side, then invert both sides.

Hint 31 | Problem 31 – Gravitational Inverse-Square Law

Replace r with $2r$. Notice that $(2r)^2 = 4r^2$, so the new force is $1/4$ of the original.

Hint 32 | Problem 32 – Conservation of Energy Speed

Cancel mass m from both sides, multiply by 2, and take the square root.

Hint 33 | Problem 33 – Kinematic Distance Formula

Subtract v_0^2 from both sides, then divide by $2a$.

Hint 34 | Problem 34 – Spring Potential Energy Stretch

Multiply by 2, divide by k , and take the positive square root.

Hint 35 | Problem 35 – Thin Lens Equation Isolation

Subtract $\frac{1}{d_o}$ from both sides, combine into a single fraction, and invert to find d_i .

Hint 36 | Problem 36 – Centripetal Acceleration and Frequency

Square $(2\pi r f)$ to get $4\pi^2 r^2 f^2$, then divide by r .

Hint 37 | Problem 37 – Atwood Machine Acceleration

Add the two equations together so T cancels out, then factor out a .

Hint 38 | Problem 38 – Pendulum Period Scaling

Substitute $4L$ in place of L and pull $\sqrt{4} = 2$ out of the square root.

Hint 39 | Problem 39 – Simple Binomial Approximation

Rewrite $\frac{1}{1+x}$ as $(1+x)^{-1}$ and apply the formula with $n = -1$.

Hint 40 | Problem 40 – Relativistic Kinetic Energy Limit

Substitute $y = \frac{v^2}{c^2}$ into $\gamma \approx 1 + \frac{1}{2} \frac{v^2}{c^2}$, then expand $K = (\gamma - 1)mc^2$.